My Assumptions for Algal Bioreactor Conceptual Design

Here are the key parameters and values assumed for the conceptual design and performance calculations of the *Chlorella vulgaris* photobioreactor.

1. **Project Scale (Population Served):** Approximately 4 small homes, which equates to 16 people in total.
   Rationale: This parameter defines the target application size for the decentralized system.

2. **Daily Wastewater Flow Rate (Q):** 2,400 Liters/day.
   Rationale: This is based on a common average estimate of 150 Liters per person per day for wastewater generation. *(Wastewater Produce | Wastewater Management | Eco-Septic, 2021)*

3. **Influent Phosphorus Concentration (Pin):** 7 mg/L.
   Rationale: Represents a typical concentration of phosphorus found in raw domestic wastewater influencers. *(Sewage Parameters 4 Part 1: Phosphorus (P), n.d.)*

4. **Influent Nitrogen Concentration (Nin):** 40 mg/L.
   Rationale: Chosen from a common range of 20-80 mg/L for nitrogen concentration in influent wastewater. *(Qin et al., 2023)*

5. **Target Nutrient Removal Efficiency (N & P):** 95%.
   Rationale: This was your initial assumed target removal efficiency for both phosphorus and nitrogen.

6. **Hydraulic Retention Time (HRT):** 6.5 days.
   Rationale: This average was chosen from a target range of 6-7 days to ensure sufficient contact time for effective nitrogen removal.

7. **Phosphorus Uptake Rate by Algae (Chlorella vulgaris):** 45.16 mg P L^{-1} d^{-1}.
   Rationale: This specific rate represents the capacity of *Chlorella vulgaris* to take up phosphorus per liter of water per day.

8. **Algae Carbon (C) Content in Dry Biomass:** Approximately 50% (0.50 as a decimal).
   Rationale: This is a standard average for carbon content in dry microalgae biomass, crucial for linking algae growth to CO_2 capture. *(Beal et al., 2018)*

9. **Algae Nitrogen (N) Content in Dry Biomass:** 8% (0.08 as a decimal).

Rationale: This is a typical percentage of nitrogen found in the dry weight of *Chlorella vulgaris* biomass, used for biomass yield calculations based on nitrogen removal. The nitrogen content of *Chlorella vulgaris* dry biomass has been reported to be 6.62% *(Saidu et al., 2021)*

10. **Algae Phosphorus (P) Content in Dry Biomass:** 1% (0.01 as a decimal).
Rationale: This is a typical percentage of phosphorus found in the dry weight of *Chlorella vulgaris* biomass, used for biomass yield calculations based on phosphorus removal. *Solovchenko et al. (2024)* noted that microalgae generally have a phosphorus content of approximately 1.8% of their dry biomass.

References

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